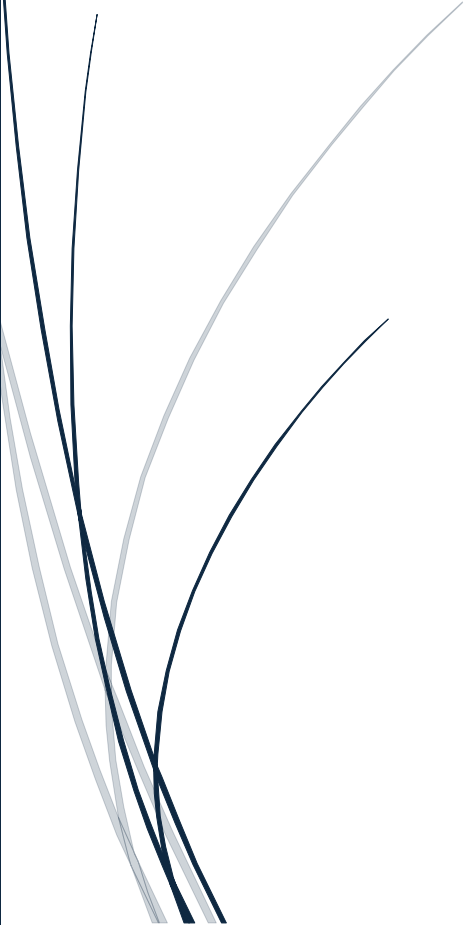


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Teen Mental Health Analysis Using Power BI

Business Intelligence Report



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1. Executive Summary

Mental health among teenagers has become an increasingly important concern for schools, healthcare providers, and policymakers. Understanding the factors associated with depression can assist organizations in developing targeted intervention strategies and promoting student well-being.

This project analysed a survey dataset containing responses from **1,200 teenagers aged 13 to 19 years**. The objective was to identify patterns associated with depression status by examining demographic information, social media usage, lifestyle habits, and mental health indicators.

Using **Power BI**, the dataset was cleaned, validated, and analysed through exploratory data analysis and business-focused visualizations. An interactive dashboard was developed to communicate the findings.

The analysis revealed that teenagers classified as depressed generally spent more time on social media, slept fewer hours, and reported significantly higher stress and anxiety levels than teenagers who were not classified as depressed. Exercise levels and study hours showed little difference between the two groups. These findings provide valuable insights that can support future mental health initiatives.

2. Business Problem

MindWell Youth Research Institute collected survey data to better understand teenage mental health. Although the data had been collected successfully, management lacked meaningful insights into how depression status differed across demographic groups and lifestyle behaviours.

Without these insights, it was difficult to determine which factors should receive greater attention when designing mental health awareness campaigns and school intervention programmes.

The purpose of this analysis was therefore to transform the raw survey data into actionable business insights that support informed decision-making.

3. Project Objectives

The objectives of this project were to:

- Inspect and clean the dataset.
- Validate data quality before analysis.
- Explore demographic characteristics of the surveyed teenagers.
- Compare lifestyle habits between teenagers classified as depressed and those who were not.
- Identify patterns associated with depression status.
- Build an interactive Power BI dashboard.
- Provide recommendations supported by the analysis.

4. Dataset Overview

The dataset consists of survey responses from **1,200 teenagers** between the ages of **13 and 19 years**.

The variables included in the dataset are grouped into the following categories.

Demographic Variables

- Age
- Gender

Social Media Variables

- Daily Social Media Hours

- Platform Usage
- Screen Time Before Sleep
- Addiction Level

Lifestyle Variables

- Sleep Hours
- Physical Activity
- Academic Performance (Study Hours)
- Social Interaction Level

Mental Health Variables

- Stress Level
- Anxiety Level

Target Variable

- Depression Status
 - ❖ 0 = Not Depressed
 - ❖ 1 = Depressed

The primary objective of the analysis was to compare teenagers classified as depressed with those who were not.

5. Data Preparation and Cleaning

Before analysis, the dataset was imported into Power BI and inspected using Power Query.

The following quality checks were completed:

- Imported the dataset into Power BI Desktop.
- Verified data types for all variables.
- Checked for missing values.
- Checked for duplicate records.
- Reviewed numerical ranges.
- Confirmed consistency of categorical values.
- Renamed columns for improved readability.

No significant data quality issues were identified. The dataset was therefore considered suitable for further analysis.

6. Exploratory Data Analysis

The exploratory data analysis provided an overview of the surveyed population before comparing depression groups.

The dashboard included:

- Total number of teenagers.
- Depression rate.
- Gender distribution.
- Age distribution.
- Social media platform usage.
- Daily social media usage.
- Sleep hours.
- Physical activity.
- Study hours.
- Stress levels.
- Anxiety levels.
- Depression status.

These visualizations provided management with an overall understanding of the survey population before more detailed comparisons were performed.

7. Business Analysis

Question 1

Do teenagers classified as depressed spend more time on social media?

Teenagers classified as depressed spent an average of **6.7 hours** per day on social media compared with **4.5 hours** among teenagers who were not classified as depressed.

This suggests that higher social media usage is associated with depression status within this dataset.

Question 2

Which social media platform has the highest proportion of teenagers classified as depressed?

The analysis showed the following depression rates:

Platform	Depression Rate
TikTok	3.02%
Instagram	2.43%
Both Platforms	2.30%

TikTok users had the highest proportion of teenagers classified as depressed. However, the differences between platforms were relatively small and should be interpreted as associations rather than causal relationships.

Question 3

Do teenagers classified as depressed sleep fewer hours?

Teenagers classified as depressed slept an average of **4.8 hours** per night compared with **6.5 hours** among teenagers who were not classified as depressed.

This was one of the strongest lifestyle differences identified during the analysis.

Question 4

Does physical activity differ between the two groups?

Teenagers classified as depressed reported an average of **0.95 hours** of daily physical activity compared with **1.02 hours** among teenagers who were not classified as depressed.

The difference was relatively small, suggesting that physical activity was not a major differentiating factor in this dataset.

Question 5

How do stress levels differ?

Teenagers classified as depressed reported an average stress level of **8 out of 10**, whereas teenagers who were not classified as depressed reported an average stress level of **5 out of 10**.

Stress level was one of the strongest factors associated with depression status.

Question 6

How do anxiety levels differ?

Teenagers classified as depressed reported an average anxiety level of **9 out of 10** compared with **6 out of 10** among teenagers who were not classified as depressed.

The results indicate a strong association between higher anxiety levels and depression status.

Question 7

How does social media addiction differ?

Unexpectedly, teenagers who were not classified as depressed reported a slightly higher average addiction level (**6 out of 10**) than teenagers classified as depressed (**5 out of 10**).

This suggests that addiction level alone is not a reliable indicator of depression within this dataset.

Question 8

Does academic performance differ?

Both groups reported an average of **3 hours** of daily study.

The analysis found no meaningful difference in study hours between teenagers classified as depressed and those who were not.

Question 9

Which gender had the higher depression rate?

The depression rates were calculated as follows:

Gender Depression Rate

Male 2.28%

Female 2.91%

Female teenagers had a slightly higher depression rate than male teenagers.

Question 10

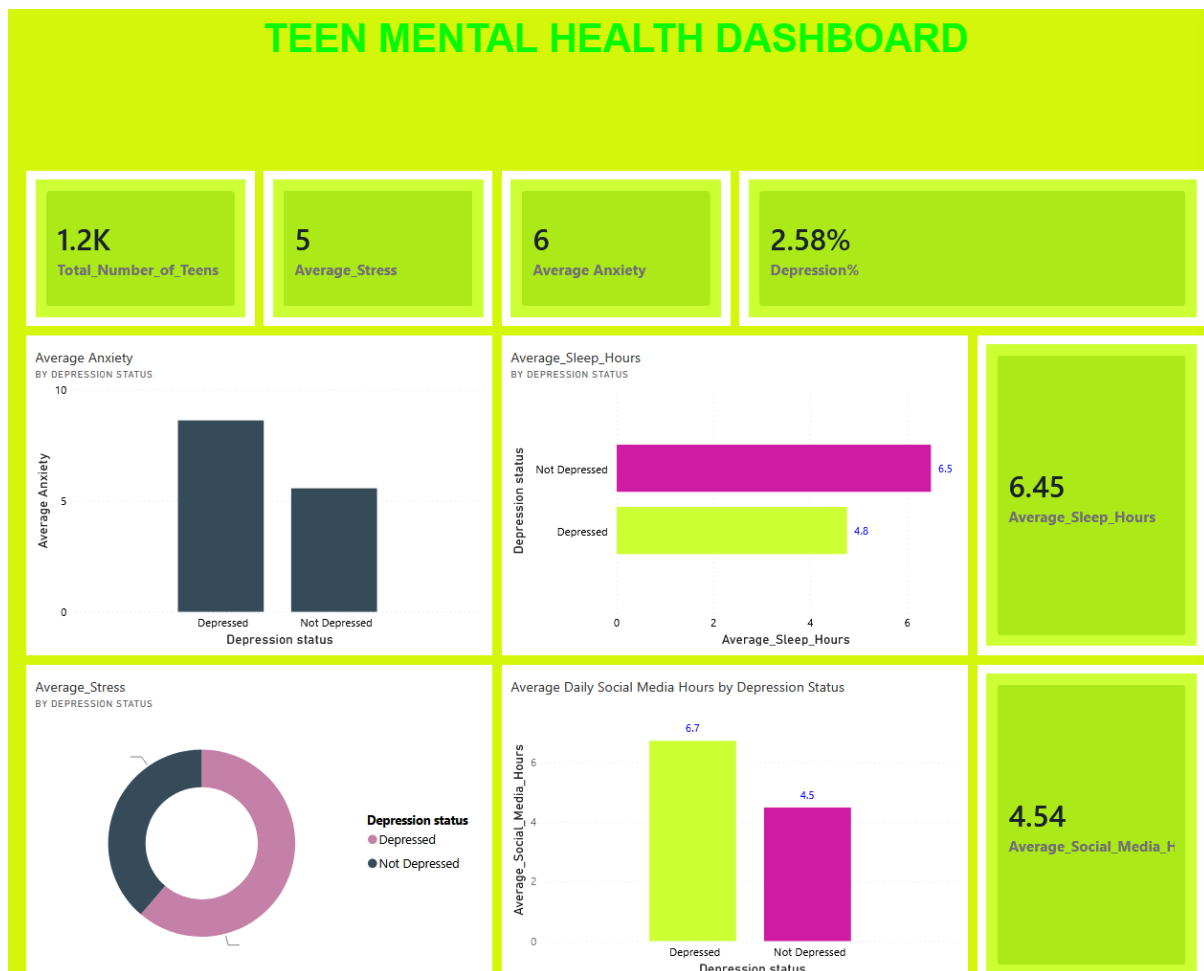
Profile of Teenagers Classified as Depressed

The typical teenager classified as depressed in this dataset had the following characteristics:

Characteristic	Value
Average Age	16 years
Most Common Gender	Female
Average Daily Social Media Use	6.7 hours
Most Common Platform	TikTok
Average Sleep	4.8 hours
Average Study Time	3 hours
Average Physical Activity	0.95 hours
Average Stress Level	8/10
Average Anxiety Level	9/10
Average Addiction Level	5/10
Social Interaction Level	Medium

8. Dashboard Overview

An interactive Power BI dashboard was developed to present the analysis in a clear and user-friendly format.

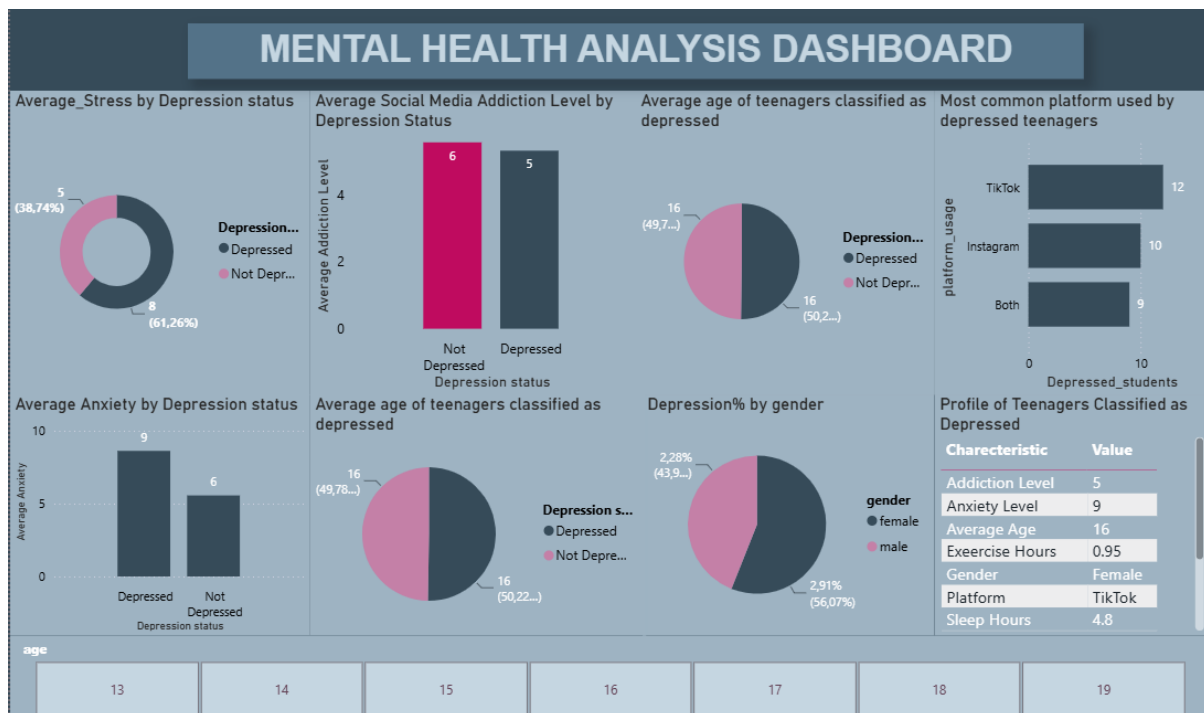


The dashboard contains:

- KPI cards displaying key metrics.
- Demographic analysis.
- Lifestyle comparisons.
- Mental health comparisons.
- Social media analysis.
- Interactive slicers allowing users to filter the report by gender, age, and platform usage.

The dashboard enables management to quickly identify trends and compare groups.





9. Key Findings

The analysis identified several important findings:

- Teenagers classified as depressed spent **2.2 more hours per day** on social media.
- They slept **1.7 fewer hours** each night.
- Stress levels were significantly higher (**8 vs. 5**).
- Anxiety levels were significantly higher (**9 vs. 6**).
- Physical activity showed only a minor difference between the two groups.
- Study hours were identical for both groups.
- Female teenagers had a slightly higher depression rate than male teenagers.
- TikTok users had the highest proportion of teenagers classified as depressed among the platform categories.

10. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Promote healthy social media habits through digital wellness education.

2. Encourage teenagers to maintain healthy sleep routines by providing education on sleep hygiene and reducing screen time before bedtime.
3. Introduce stress management programmes in schools, including mindfulness activities, resilience training, and counselling services.
4. Increase access to mental health support and anxiety screening for students who may require early intervention.
5. Continue promoting regular physical activity as part of a healthy lifestyle, while recognising that it showed only a small difference in this dataset.
6. Use multiple indicators, including stress, anxiety, sleep, and social media use, when identifying students who may benefit from mental health support.
7. Continue collecting and analysing mental health data to monitor trends and evaluate future intervention programmes.

11. Limitations

This analysis is based on survey responses from **1,200 teenagers**, of whom only **31 were classified as depressed (2.58%)**. As a result, the dataset is imbalanced, and findings should be interpreted carefully.

Furthermore, this study is based on observational survey data. While the analysis identifies associations between variables and depression status, it does not establish causal relationships.

Future research using larger datasets and additional variables could provide more comprehensive insights

12. Conclusion

This project successfully transformed raw survey data into meaningful business insights using Power BI.

The findings indicate that teenagers classified as depressed generally reported higher social media use, reduced sleep, and substantially higher stress and anxiety levels compared with teenagers who were not classified as depressed. Other variables, such as physical activity and study hours, showed minimal differences.

The interactive Power BI report provides stakeholders with an accessible tool for exploring these findings and supports evidence-based decision-making. By applying business intelligence techniques and data visualization, this project demonstrates how data analytics can contribute to understanding important public health issues and guide future mental health initiatives.